Environmental toxicology

It is a multidisciplinary field of science concerned with the study **of** harmful effects of various chemical, biological and physical agents on living organisms. Humans, animals, and plants are increasingly being exposed to chemicals in the environment. The ever-increasing use of chemicals in industries has also resulted in further pollution of the environment.

As toxic chemicals are widespread in the environment, there is a potential for these chemicals to cause significant damage and harmful effects on human health.

- Almost any chemical can be harmful depending on the dose (the amount of chemical and amount of time a person is exposed, in addition to how often a person is exposed) and route of exposure (whether the chemical is eaten, breathed in, or touches your skin). **Acute** (short-term) exposure and **chronic** (long-term) exposures can have different effects on a person's health.

Environmental hazard

A hazard is a situation that poses a level of threat to life, health, property, or the environment. An environmental hazard is an event that has the potential to threaten the surrounding natural environment or adversely affect people's health, including pollution and natural disasters.

- **Types of environmental Hazards:** hazards are classified into five
- **Physical**: occur naturally or due to human factors in our environment, e.g., earthquakes, floods, volcanoes, drought, and fire outbreaks.
- **Cause**: due to deforestation, channelizing rivers (flooding), and deforesting slopes (landslides)
- **Reduce:** by better environmental choices, planting trees, and avoiding excess cultivation of land



Safety and Prevention

Ways to lower **morbidity rates** include increasing **screenings and early diagnosis, which would lessen the length and impact of the disease on a person's quality of life.**

This would also lower complications and the mortality rates of certain diseases because early treatment is often the most effective.

Other ways to compress morbidity are through **education and access to preventive healthcare**. For example, a model to lower morbidity for pregnant women involves access to safe abortions, prenatal care throughout pregnancy, and postpartum care, along with family planning education throughout.

As people live longer, the focus on lowering morbidity is on education throughout life to create healthy habits and monitor health outcomes. A critical step is to receive regular check-ups to promote lifelong health before any signs of disease occur.

Emerging disease due to environmental changes

- Environmental changes have a huge impact on the emergence of certain infectious diseases, mostly in countries with high biodiversity and serious unresolved environmental, social, and economic issues.
- **An extensive review revealed a relationship between infectious disease outbreaks and environmental and climate change (heat wave, drought, flood, increased temp, high rainfall, etc) or environmental changes such as habitat fragmentation, deforestation, urbanization, and bush meat consumption.**

- According to the World Health Organization, **environmental threats to human health at global and regional levels include: "climate change, stratospheric ozone depletion, changes in ecosystems due to loss of biodiversity, changes in hydrological systems and supplies of freshwater, land degradation, urbanization, and stresses on food-producing systems" (WHO 2017).**
- Rapid social and environmental changes are occurring globally, affecting both low-income countries and the largest advanced economies.
- Travel and transportation are the greatest increase in the risks for the rapid spread of infectious diseases (Jaffry et al. 2009). Particularly in the tropics, where there is high biodiversity, ecological change is greatest, making these regions potential hotspots for the emergence of new pathogens affecting human, wildlife, and domestic animal health (Jones et al. 2008; Wilcox and Gubler 2005).

- These potentially lead to changes in microbial ecological diseases affecting human health. Sociodemographic **factors such as an increase in population density, falling living standards, the decline of infrastructure, human travel, conflicts, social instability, and the killing of wild animals for meat all lead to an increase in emerging ecological diseases, which facilitate infections in humans**. There are pathogens (disease-causing organisms) whose emergence is a result of deliberate human action.
- A reasonable public health response towards addressing emerging ecological disease problems generally aims at addressing **the fundamental factors that promote the occurrence and persistence of these diseases**, while embarking on **appropriate control measures**. WHO (World Health Organization), therefore, supports **advocacy, awareness, and pathogenesis studies, development and deployment of diagnostic tools, therapeutic drugs, and vaccines** as the pillars of **public health response**

Toxic chemicals

The U.S. Environmental Protection Agency (EPA) defines a **toxic chemical as any substance which may be harmful to the environment or hazardous to your health if inhaled, ingested or absorbed through the skin.**

- **Toxic Chemicals in Your Home**
- Many useful household products contain toxic chemicals. Common examples include:
- Drain cleaner
- Laundry detergent
- Furniture polish
- Gasoline
- Pesticides
- Ammonia
- Toilet bowl cleaner
- Motor oil
- Rubbing alcohol
- Bleach
- Battery acid. While these chemicals may be useful and even necessary, it is important to remember they should be used and disposed according to the instructions on the packaging.

Types of Toxins

- Toxins may be categorized into four groups. It's possible for a substance to belong to more than one group.
- **Chemical Toxicants** - Chemical toxins include both **inorganic substances**, such as mercury, ammonia, hydrogen sulfide, all metals, and most elements. , and **organic compounds**, such as methyl alcohol.
- **Biological Toxins** - Many organisms secrete toxic compounds. Some sources consider pathogenic organisms to be toxins. A good example of a biological toxin is tetanus.
- **Physical Toxicants** - These are substances due to their physical nature that interfere with biological processes. Examples include coal dust, asbestos fibres (naturally occurring fibrous silicate mineral) which can ultimately be fatal if inhaled.
- **Radiation** - Radiation has a toxic effect on many organisms. Examples include gamma radiation and microwaves.

Here are some tips for minimizing your exposure

- Consider choosing products with BPA- and phthalate-free packaging.
- Drink tap water that has been filtered.
- Avoid microwaving plastics.
- Avoid personal care products that list “fragrance” as an ingredient, as phthalates often hide in fragrances.
- Choose products labeled “phthalate-free,” “paraben-free,” or “BPA-free.”
- Avoid unnecessary exposure to industrial chemicals and pesticides.





Diet

- In nutrition, **diet** is the sum of food consumed by a person or other organism. The word diet often implies the use of specific intake of nutrition for health or weight-management reasons (with the two often being related). Although humans are omnivores (ability to eat and survive on both plant and animal matter, obtaining energy and nutrients from plant and animal, each culture and each person holds some food preferences or some food taboos. This may be due to personal tastes or ethical reasons. Individual dietary choices may be more or less healthy.
- Complete nutrition requires ingestion and absorption of vitamins, minerals, essential amino acids from protein, and essential fatty acids from fat-containing food, also food energy in the form of carbohydrate, protein, and fat. Dietary habits and choices play a significant role in the quality of life, health and longevity (life expectancy).